Trainings

General Information

with Electronically Actuated Injector

This procedure is for engines with electronically actuated injectors **only**.

Use the following procedure for engines with mechanically actuated injectors. Refer to Procedure 009-001 in Section 9. (/qs3/pubsys2/xml/en/procedures/20/20-009-001-tr.html)

The fuel pump drive on QSK19 engines with electronically actuated injectors contain an oil feed and drain drillings which are larger and in a different location. The QSK19 with electronically actuated injectors does **not** use the fuel pump drive to drive the air compressor.

Preparatory Steps

with Electronically Actuated Injector

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html)
- Remove the accessory drive pulley, if applicable. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/20/20-009-004.html)
- Remove the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-003-tr.html)

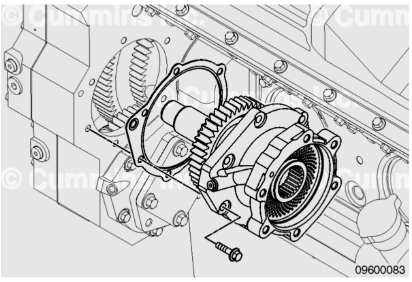
Remove

with Electronically Actuated Injector

Remove the six capscrews.

Remove the accessory drive assembly.

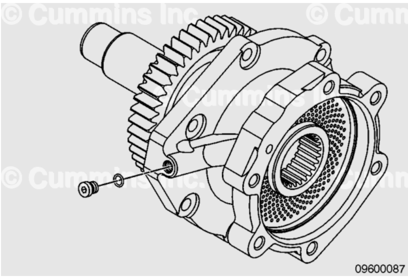
Remove and inspect the gasket.



Disassemble

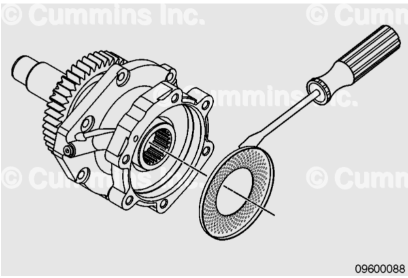
with Electronically Actuated Injector

Remove the straight thread o-ring plug from the fuel pump drive.

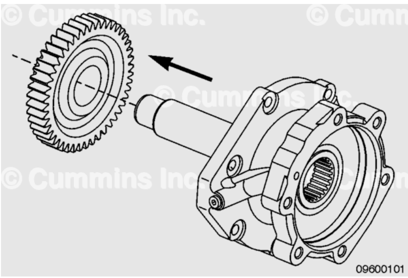


Remove the mesh screen with a small flat tip screwdriver.

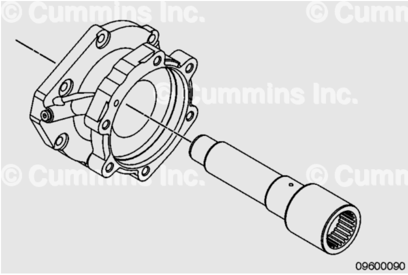
If debris is found in the screen, check the fuel pump for proper operation.



Use a press to remove the gear from the shaft.

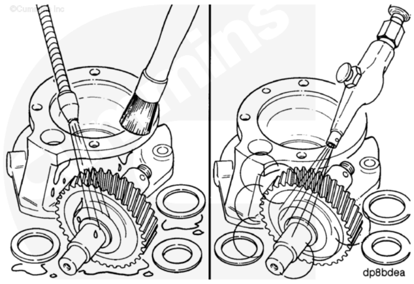


Remove the shaft from the housing.



Clean and Inspect for Reuse

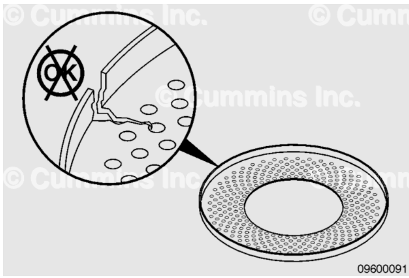
with Electronically Actuated Injector



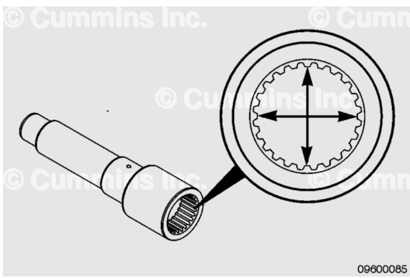
⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean the components with solvent before inspection.



Check the screen for cracks or other damage. If any damage is found, the screen **must** be replaced.



Check the inner diameter of the shaft spline for damage.

If damage is found, the shaft **must** be replaced.

Measure the inside diameter of the shaft.

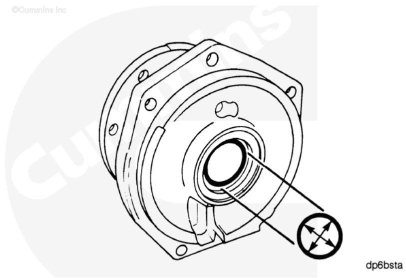
Fuel Pump Drive Spline Inside Diameter		
mm		in
46.00	MIN	1.811
46.29	MAX	1.822

If the inner diameter of the shaft is **not** within specification, the shaft **must** be replaced.

Measure the inside diameter of the bushings.

Fuel Pump Drive Bushing Inside Diameter		
mm		in
50.00	MIN	1.968
50.03	MAX	1.970

If the inside diameter of either bushing is **not** within specification, both bushings **must** be replaced.

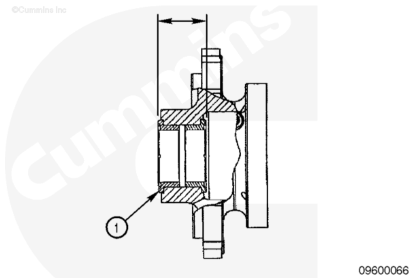


Check the grooved surfaces of the bushings (1) for damage.

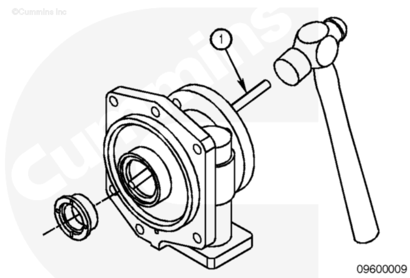
Measure the distance from the outside of the thrust faces of the two bushings.

Fuel Pump Drive Bushing Thrust Face Distance		
mm		in
45.375	MIN	1.786
45.450	MAX	1.789

If the distance between the two thrust faces is **not** within specification, both bushings **must** be replaced.

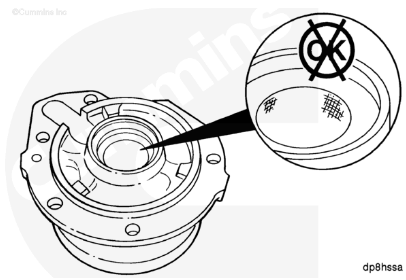


Remove the flanged bushing from the drive housing. Use a brass drift (1), taking care **not** to damage the bushing bore.



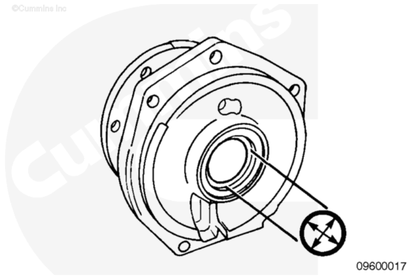
If the bushing was removed, inspect the housing for scoring and other damage.

Replace the housing, if damaged.



If the bushing was removed, measure the inside diameter of the bushing bore.

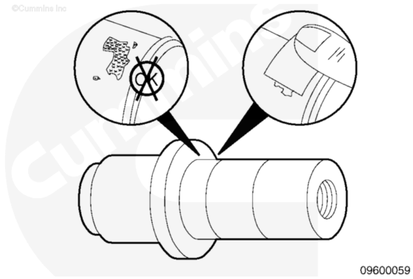
Fuel Pump Drive Bushing Bore Inside Diameter		
mm		in
60.00	MIN	2.362
60.05	MAX	2.364



Replace the housing if the bushing bore is **not** within specifications.

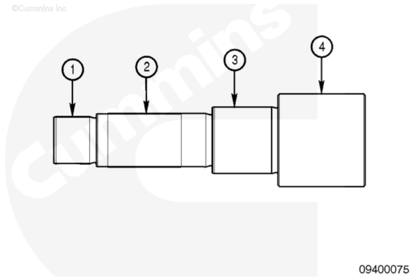
Inspect the shaft in the gear fit area for fretting or burrs.

Clean burrs with emery cloth.



Measure the shaft outside diameter.

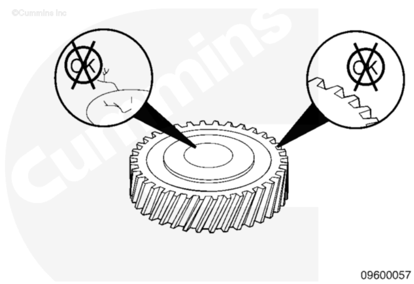
Fuel Pump Drive Shaft Outside Diameter			
	mm		in
(1)	34.963	MIN	1.376
	34.975	MAX	1.377
(2)	39.662	MIN	1.561
	39.674	MAX	1.562
(3)	49.820	MIN	1.961
	49.832	MAX	1.962
(4)	68.75	MIN	2.706
	69.25	MAX	2.726



If the outside diameter is **not** within specification, the shaft **must** be replaced.

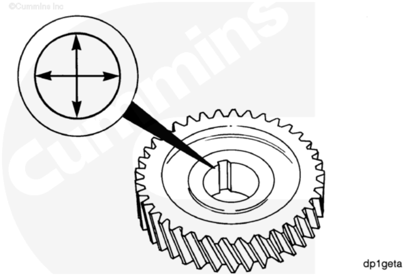
Check the gear teeth and gear bore for damage.

Replace the gear, if damaged.



Measure the gear inside diameter.

Fuel Pump Drive Gear Inside Diameter		
mm		in
39.726	MIN	1.564
39.751	MAX	1.565

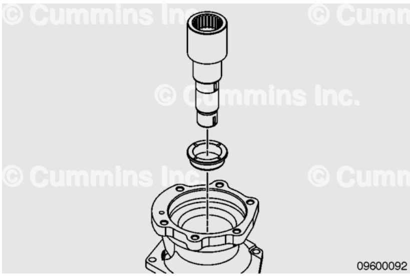


If the gear is **not** within specification, the gear **must** be replaced.

Assemble

with Electronically Actuated Injector

If the bushings were removed, use the shaft as a mandrel and press one bushing into the housing until the underside of the bushing head is flush with the counterbore in the housing. Turn the housing over and press the remaining bushing into place.

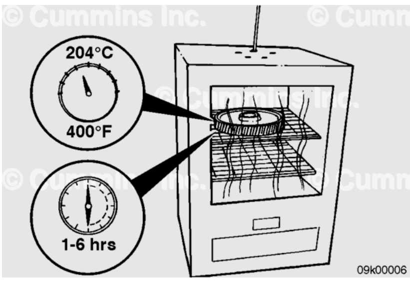


⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

⚠ CAUTION ⚠

Do not exceed the specified time or temperature. Damage to the gear teeth will result.

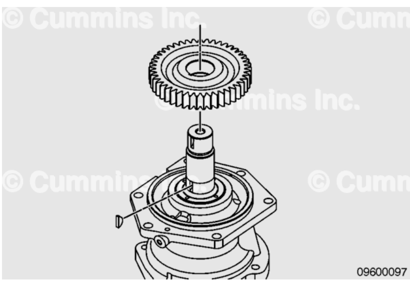


If an adequate press is **not** available, an oven can be used.

Heat the gear 204°C [400°F] for **not** less than 1 hour, and **not** more than 6 hours.

⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.



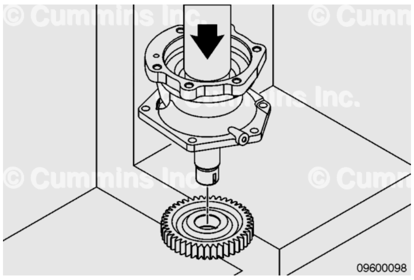
Support the shaft and housing.

Slide the gear onto the shaft until the shoulder of the shaft touches the gear.

If an arbor press is available, support the gear.

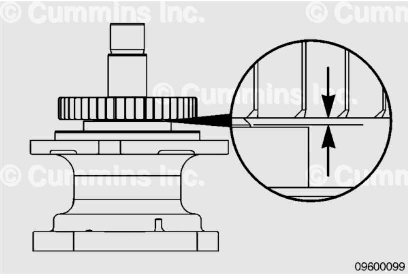
Lubricate the shaft with clean engine oil.

Press the shaft through the gear until the shoulder of the shaft touches the gear.



⚠ CAUTION ⚠

If the heat method was used to install the gear, allow the gear to air cool. Do not use water or oil to reduce cooling time. Damage to the gear can result.

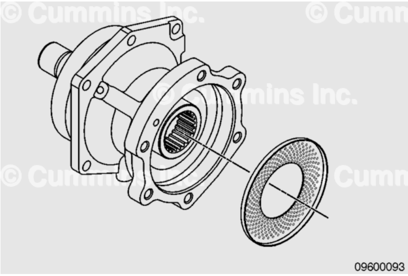


Use a feeler gauge to measure the distance between the shoulder of the shaft and the gear.

Fuel Pump Drive Gear to Shaft Distance		
mm		in
0.05	MAX	0.002

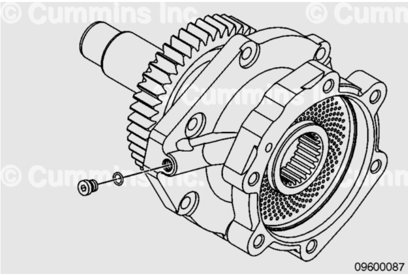
If the distance between the gear and the shaft is **not** within specification, press the gear on until the specification is met.

Install the screen into the housing by pushing the screen in by hand.



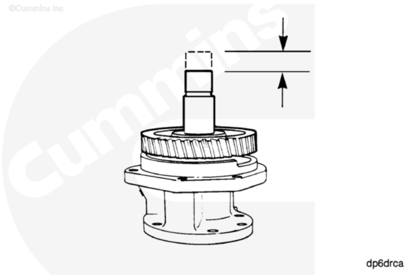
Install the straight thread o-ring plug.

Torque Value: 6 n•m [35 in-lb]



Rotate the shaft, checking for correct assembly.

Use a dial indicator to measure the fuel pump drive shaft end clearance.



Fuel Pump Drive Shaft End Clearance		
mm		in
0.13	MIN	0.0051
0.27	MAX	0.0106

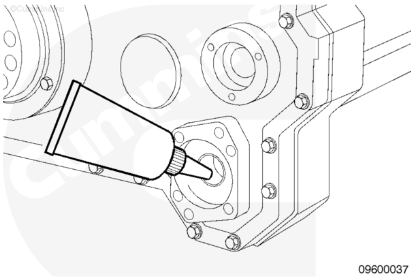
If the end clearance is **not** within specifications, make sure the coupling is positioned tightly against the clamping washer.

Install

with Electronically Actuated Injector

⚠ CAUTION ⚠

Do not use sealants on the gasket. The gasket is manufactured from material that sealants will damage.

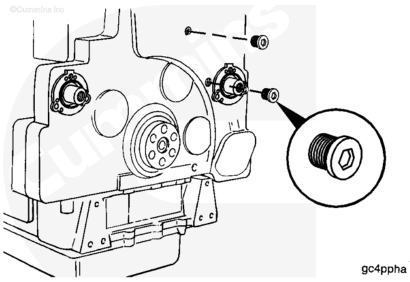


Lubricate the bushing in the front gear housing cover with Lubriplate™ 105.

Install the gasket onto the pilot of the drive housing cover.

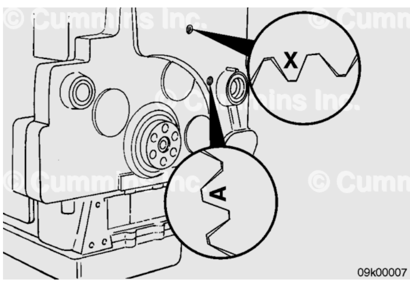
Remove the two straight threaded o-ring plugs from the timing holes in the front cover.

Check the index mark alignment.



⚠ CAUTION ⚠

Make sure the "X" mark on the camshaft gear is centered in the upper timing plug hole before centering the "A" mark on the accessory drive gear. The marks on the camshaft idler



gear do not have to align with either of the marks on the camshaft gear or accessory drive gear as the camshaft idler gear alignment does not affect timing on MCRS engines.

Make sure the "X" mark on the camshaft gear is centered in the upper timing plug hole.

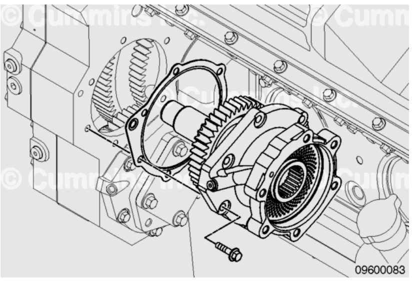
Install the accessory drive so that the "A" on the accessory drive gear is centered in the lower timing plug hole.

Note : For MCRS engines, it is only necessary to center the "X" mark on the camshaft gear in the upper timing hole and center the "A" mark on the accessory drive gear in the lower timing hole. The marks on the camshaft idler gear do not have to be aligned for accessory drive timing.

Install the fuel pump drive assembly, turning the shaft as necessary to engage the gears.

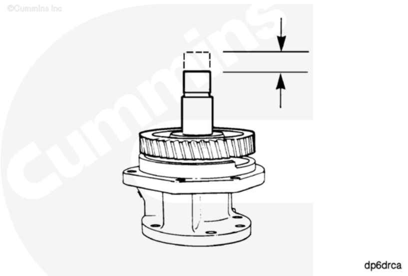
Install and tighten the six capscrews.

Torque Value: 45 n•m [33 ft-lb]



With the fuel pump drive installed, measure the end clearance.

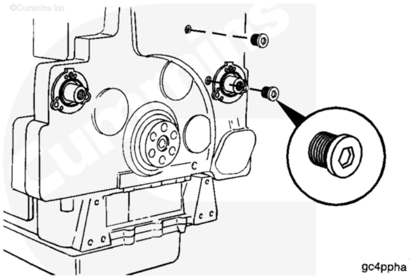
Fuel Pump Drive End Clearance		
mm		in
0.13	MIN	0.0051
0.27	MAX	0.0106



Install straight threaded o-ring plugs.

Tighten the plugs.

Torque Value: 25 n•m [221 in-lb]



Finishing Steps

with Electronically Actuated Injector

- Install the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-003-tr.html)
- Install the accessory drive pulley, if applicable. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/20/20-009-004.html)
- Install the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine until the coolant temperature reaches 71°C [160°F]. Check for leaks.

Last Modified: 20-Aug-2015